

# RAMIFICATION OF PRIMES: A PRESENTATION FOR MATH 8600: COMMUTATIVE ALGEBRA

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## 1. INTRODUCTION

Let  $K|\mathbb{Q}$  be a finite field extension with  $[K : \mathbb{Q}] = n$ . Then, we may consider the integral closure of  $\mathbb{Z}$  in  $K$ , say  $\mathcal{O}_K$ . Thus, we have the following setup.

$$\begin{array}{ccc} K & \longleftrightarrow & \mathcal{O}_K \\ | & & | \\ \mathbb{Q} & \longleftrightarrow & \mathbb{Z} \end{array}$$

where  $\mathcal{O}_K|\mathbb{Z}$  is an integral ring extension. Now, recall the following facts.

**1.1. Proposition.** *Given the setup above*

- (a)  $\mathcal{O}_K$  is a Dedekind domain.
- (b) Given a prime  $p \in \mathbb{Z}$ , the ideal  $(p) = p\mathcal{O}_K \trianglelefteq \mathcal{O}_K$  has a unique decomposition

$$(p) = \prod_{i=1}^g P_i^{e_i}$$

for prime ideals  $P_i \trianglelefteq \mathcal{O}_K$  and  $e_i \in \mathbb{N}$ .

- (c)  $\mathcal{O}_K$  is a finitely-generated, free  $\mathbb{Z}$ -module, say

$$\mathcal{O}_K \cong \mathbb{Z}\alpha_1 \oplus \cdots \oplus \mathbb{Z}\alpha_n \text{ as a } \mathbb{Z}\text{-module.}$$

Thus,  $\mathcal{O}_K/p\mathcal{O}_K$  is a finitely-generated  $\mathbb{Z}/p\mathbb{Z}$ -module, that is

$$\mathcal{O}_K/p\mathcal{O}_K \cong (\mathbb{Z}/p\mathbb{Z})\overline{\alpha_1} \oplus \cdots \oplus (\mathbb{Z}/p\mathbb{Z})\overline{\alpha_n}$$

Furthermore, by the Chinese Remainder Theorem,

$$\mathcal{O}_K/p\mathcal{O}_K \cong \mathcal{O}_K/P_1^{e_1} \times \cdots \times \mathcal{O}_K/P_g^{e_g}$$

so each  $\mathcal{O}_K/P_i^{e_i}$  is an  $\mathbb{F}_p$ -vector space, and in fact, an  $\mathbb{F}_p$ -algebra since  $p \in P_i^{e_i}$ .

This leads us to the following definition:

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**1.2. Definition.** We say a prime  $p \in \mathbb{Z}$  is *ramified* in  $\mathcal{O}_K$  if

$$p\mathcal{O}_K = \prod_{i=1}^g P_i^{e_i}$$

has some  $e_i > 1$  for prime ideals  $P_i \subseteq \mathcal{O}_K$ . If every  $e_i = 1$ , then  $p$  is *unramified* in  $\mathcal{O}_K$ .

**1.3. Example.** Consider  $2 \in \mathbb{Z}[i]$ . Then, since

$$-i(1+i)(1+i) = -i(1+2i-1) = -i2i = 2,$$

we have that  $(2) \subseteq (1+i)^2$ . Furthermore, since  $(1+i)$  is prime in  $\mathbb{Z}[i]$  using norm arguments, and  $(2)$  has norm 4, it must be that  $(2) = (1+i)^2$ . Therefore, 2 ramifies in  $\mathbb{Z}[i]$ .

We wish to come up with some method to determine when a prime will ramify in  $\mathcal{O}_K$ . One such characterization uses the notion of the “discriminant.”

**1.4. Definition.** Let  $V$  be an  $m$ -dimensional vector space over  $K$ . Then, given a symmetric bilinear form  $b: V \times V \rightarrow K$  and  $\{\omega_1, \dots, \omega_m\}$  a basis of  $V$ , we define

$$\text{disc}(b; \omega_1, \dots, \omega_m) := \det(b(\omega_i, \omega_j))_{1 \leq i, j \leq m}$$

**1.5. Proposition.** Given another  $K$ -basis of  $V$  as above, say  $\{\omega'_1, \dots, \omega'_m\}$  such that

$$M \begin{pmatrix} \omega_1 \\ \vdots \\ \omega_m \end{pmatrix} = \begin{pmatrix} \omega'_1 \\ \vdots \\ \omega'_m \end{pmatrix}$$

we get that

$$\text{disc}(b; \omega'_1, \dots, \omega'_m) = (\det M)^2 \text{disc}(b; \omega_1, \dots, \omega_m)$$

*Proof.* Consider that if

$$B = (b(\omega_i, \omega_j))_{1 \leq i, j \leq m}, \quad B' = (b(\omega'_i, \omega'_j))_{1 \leq i, j \leq m}$$

then,

$$B'_{i,j} = b(\omega'_i, \omega'_j) = b\left(\sum_{k=1}^n m_{k,i} \omega_k, \sum_{\ell=1}^n m_{\ell,j} \omega_\ell\right) = \sum_{k=1}^n \sum_{\ell=1}^n m_{i,k} b(\omega_k, \omega_\ell) m_{j,\ell} = (MBM^t)_{i,j}$$

and so  $B' = MBM^t$ . Then the result is obtained by taking the determinant of both sides.  $\square$

**1.6. Definition.** Let  $K$  be a field and let  $A$  be a finite-dimensional  $K$ -algebra with basis  $\{x_1, \dots, x_n\}$ . Then,

- (a) The *trace*  $\text{Tr}_{A|K}(z) := \text{tr } m_z$  where, if

$$zx_i = \sum_{j=1}^n a_{i,j} x_j, \quad a_{i,j} \in K$$

then  $m_z = (a_{i,j})_{1 \leq i,j \leq n}$ . Note that this is independent of choice of basis since a different choice will give a matrix  $m'_z$  that is conjugate to  $m_z$ , which will not change the trace.

- (b) The *trace form*  $T: A \times A \rightarrow K$  is given by

$$T(x, y) = \text{Tr}_{A|K}(xy)$$

Since we are in a commutative ring, the form is symmetric. Since matrix trace is bilinear, then so is the trace form.

- (c) The *discriminant* of  $A$  is

$$\text{disc}(A) := \text{disc}(T; x_1, \dots, x_n)$$

**1.7. Remark.** Consider the case that  $K|\mathbb{Q}$  is a finite separable field extension with  $\mathcal{O}_K \subseteq K$  the integral closure of  $\mathbb{Z}$  in  $K$ .

- (a) Then, the discriminant is independent of choice of integral basis since, given another integral basis  $\{x'_1, \dots, x'_n\}$ , we have

$$\text{disc}(T; x'_1, \dots, x'_n) = (\det M)^2 \text{disc}(T; x_1, \dots, x_n)$$

However,  $M$  is an invertible matrix with entries in  $\mathbb{Z}$ , so it must be that  $\det M = \pm 1 \implies (\det M)^2 = 1$ .

- (b) Note  $\text{disc}(K)$  is always an integer because  $\text{Tr}_{K|\mathbb{Q}}(\mathcal{O}_K) \subseteq \mathbb{Z}$ .

**1.8. Example.** Consider the field extension  $\mathbb{Q}(i)|\mathbb{Q}$ . Then, if we take integral basis  $\{1, i\}$ , we get

$$m_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, m_i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \text{ and } m_{-1} = -m_1$$

Thus,

$$\text{Tr}(1) = 2, \text{Tr}(i) = 0, \text{Tr}(-1) = -2$$

and so

$$\text{disc}(\{1, i\}) = \det \begin{pmatrix} \text{Tr}(1) & \text{Tr}(i) \\ \text{Tr}(i) & \text{Tr}(-1) \end{pmatrix} = \det \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = -4$$

This paper seeks to prove the following useful characterization for when a prime  $p$  ramifies in  $\mathcal{O}_K$ .

**1.9. Theorem.** *A prime  $p \in \mathbb{Z}$  ramifies in  $\mathcal{O}_K$  if and only if  $p \mid \text{disc}(K)$ .*

From this result, we also have the useful corollary

**1.10. Corollary.** *Only a finite number of primes  $p \in \mathbb{Z}$  ramify in  $\mathcal{O}_K$ .*

Thus, from our running example, 2 is the only prime that ramifies in  $\mathbb{Z}[i]$ . In the next section, we will follow a synthesis of the programs by [Ash03, 4.2] and [Con] to prove this theorem.

## 2. STRUCTURE AND TRACE OF THE QUOTIENT $\mathcal{O}_K/p\mathcal{O}_K$

Using our same setup, let  $(p) = p\mathcal{O}_K = \prod_i P_i^{e_i}$  for prime ideals  $P_i \subseteq \mathcal{O}_K$  and  $e_i \in \mathbb{N}$ .

**2.1. Lemma.**  *$p$  ramifies if and only if the ring  $\mathcal{O}_K/(p)$  has nonzero nilpotent elements.*

*Proof.* •  $(\implies)$ . Let  $p$  ramify in  $\mathcal{O}_K$ . Then,  $\mathcal{O}_K/p\mathcal{O}_K \cong \mathcal{O}_K/P_1^{e_1} \times \cdots \times \mathcal{O}_K/P_n^{e_n}$  by the Chinese Remainder Theorem, where at least one  $e_i > 1$ , let us say  $e_1$ . Then, the quotient ring  $\mathcal{O}_K/P_1^{e_1}$  has a nonzero nilpotent element since, for  $x \in P_1 \setminus P_1^{e_1}$ , we get  $(x + P_1^{e_1})^{e_1} = x^{e_1} + P_1^{e_1} = P_1^{e_1}$ .  
•  $(\impliedby)$ . If  $p$  does not ramify in  $\mathcal{O}_K$ , then  $\mathcal{O}_K/p\mathcal{O}_K \cong \mathcal{O}_K/P_1 \times \cdots \times \mathcal{O}_K/P_n$ , each of which is a field since each  $P_i$  is maximal in  $\mathcal{O}_K$ . Furthermore, each of these fields is finite by Proposition 1.1(c). Thus,  $\mathcal{O}_K/p\mathcal{O}_K$  cannot have any nonzero nilpotent elements.  $\square$

We also have, as a corollary to the proof, that

**2.2. Corollary.** *If  $p$  is unramified in  $\mathcal{O}_K$ , then  $\mathcal{O}_K/p\mathcal{O}_K$  is a product of finite fields.*

This is a useful fact since

**2.3. Lemma.** *A nilpotent element has zero trace.*

*Proof.* Let  $x^n = 0$  for some  $n \in \mathbb{N}$ . Then, since  $m_{x^k} = (m_x)^k$ , it must be that  $(m_x)^n = 0$ , so  $m_x$  is a nilpotent matrix, which has trace 0 since its minimal polynomial  $\mu_{m_x}(t) \mid t^n$ . Therefore,

$$\mathrm{Tr}_{K|\mathbb{Q}}(x) = \mathrm{tr} m_x = 0$$

$\square$

And so, we get

**2.4. Lemma.** *For prime  $p \in \mathbb{Z}$ , let  $p\mathcal{O}_K = \prod_{i=1}^g P_i^{e_i}$ . For any  $e_i > 1$ ,  $\mathrm{disc}_{\mathbb{F}_p}(\mathcal{O}_K/P_i^{e_i}) = \bar{0}$ .*

*Proof.* From 1.1(c), we have that  $\mathcal{O}_K/P_i^{e_i}$  is an  $\mathbb{F}_p$ -algebra. By the above, since at least one  $e_i > 1$ ,  $p$  ramifies and so we know  $\mathcal{O}_K/P_i^{e_i}$  has a nonzero nilpotent element, say  $x$ . Then, extend  $\{x\}$  to a basis of  $\mathcal{O}_K/P_i^{e_i}$  over  $\mathbb{F}_p$ , say  $\{x, x_2, \dots, x_k\}$ . Each  $xx_i$  is nilpotent, so, for all  $i$ ,

$$\mathrm{Tr}_{\mathcal{O}_K/P_i^{e_i}|\mathbb{F}_p}(xx_i) = \bar{0}$$

and so, since the trace form matrix will have a row of all zeros, it must have determinant equal to  $\bar{0}$  and so the discriminant is 0.  $\square$

**2.5. Lemma.** *Let  $p$  is in  $\mathcal{O}_K$  be unramified, that is,  $p\mathcal{O}_K = \prod_{i=1}^g P_i$ . Then, the trace form of  $\mathcal{O}_K/P_i$  over  $\mathbb{F}_p$  is nondegenerate. Thus, given the field extension  $\mathcal{O}_K/P_i|\mathbb{F}_p$ , the discriminant*

$$\text{disc}(\mathcal{O}_K/P_i) \neq \bar{0} \in \mathbb{F}_p$$

*Proof.* By the arguments above, we already know that  $\mathcal{O}_K/P_i$  is a finite field, and since  $\mathbb{F}_p$  is perfect, we have that  $\mathcal{O}_K/P_i|\mathbb{F}_p$  is a separable field extension. Therefore, by Lemma 2.2.3 in class, it must be that the trace form is nondegenerate. Therefore, fixing an  $\mathbb{F}_p$ -basis of  $\mathcal{O}_K/P_i$ ,  $\{\omega_1, \dots, \omega_k\}$  the matrix

$$(T(\omega_i, \omega_j))_{1 \leq i, j \leq n} \text{ is invertible } \iff \det(T(\omega_i, \omega_j))_{1 \leq i, j \leq n} \neq \bar{0}$$

Therefore,  $\text{disc}(\mathcal{O}_K/P) \neq \bar{0}$ .  $\square$

### 3. DISCRIMINANT BEHAVES WELL WITH REDUCTION mod $p$ AND PRODUCTS

**3.1. Lemma.** *For an appropriate choice of bases,*

$$\text{disc}(K) \bmod p = \text{disc}(\mathcal{O}_K/p\mathcal{O}_K)$$

*Proof.* Let  $\{\alpha_1, \dots, \alpha_n\}$  be an integral basis for  $\mathcal{O}_K|\mathbb{Z}$ . Then, for  $x \in \mathcal{O}_K$ , we have  $a_{i,j} \in \mathbb{Z}$  such that

$$x\alpha_i = \sum_j a_{i,j}\alpha_j \implies x\alpha_i + p\mathcal{O}_K = \sum_j \overline{a_{i,j}}\alpha_j + p\mathcal{O}_K$$

where  $\overline{a_{i,j}} = a_{i,j} \bmod p$ . Thus,  $m_x$  with the entries reduced mod  $p$  is equal to  $m_{x+p\mathcal{O}_K}$ . Thus,

$$\text{Tr}_{\mathcal{O}_K/p\mathcal{O}_K|\mathbb{F}_p}(x+p\mathcal{O}_K) = \text{tr}(m_{x+p\mathcal{O}_K}) = \text{tr}(m_x) \bmod p = \text{Tr}_{K|\mathbb{Q}}(x) \bmod p$$

giving us that

$$(\text{Tr}_{K|\mathbb{Q}}(\alpha_i\alpha_j))_{1 \leq i, j \leq n} \bmod p = \text{Tr}_{\mathcal{O}_K/(p)|\mathbb{Z}/p\mathbb{Z}}(\overline{\alpha_i}\overline{\alpha_j})$$

and so, taking determinants of both sides gives the desired result.  $\square$

**3.2. Lemma.** *Let  $F$  be a field with  $B_1, B_2$  finitely-generated  $F$ -algebras. Then, up to appropriate choice of basis,*

$$\text{disc}(B_1 \times B_2) = \text{disc}(B_1) \text{disc}(B_2)$$

*Proof.* Let

$$B_1 = \bigoplus_{i=1}^m Fe_i, \quad B_2 = \bigoplus_{j=1}^n Ff_j$$

Then, take the standard choice of  $F$ -basis of  $B_1 \times B_2$ ,  $\{e_1, \dots, e_m, f_1, \dots, f_m\}$ . Since  $e_i f_j = 0$  in  $B_1 \times B_2$ , we get that

$$\text{disc}(B_1 \times B_2) = \det \begin{pmatrix} \text{Tr}_{B_1 \times B_2|F}(e_i e_k) & 0 \\ 0 & \text{Tr}_{B_1 \times B_2|F}(f_j f_\ell) \end{pmatrix}$$

Also, for  $x \in B_1$ , since  $xy = 0$  for all  $y \in B_2$ , we have

$$\mathrm{Tr}_{B_1 \times B_2|F}(x) = \mathrm{Tr}_{B_1|F}(x)$$

and similarly for  $y \in B_2$

$$\mathrm{Tr}_{B_1 \times B_2|F}(y) = \mathrm{Tr}_{B_2|F}(y)$$

Thus,

$$\begin{pmatrix} \mathrm{Tr}_{B_1 \times B_2|F}(e_i e_k) & 0 \\ 0 & \mathrm{Tr}_{B_1 \times B_2|F}(f_j f_\ell) \end{pmatrix} = \begin{pmatrix} \mathrm{Tr}_{B_1|F}(e_i e_k) & 0 \\ 0 & \mathrm{Tr}_{B_2|F}(f_j f_\ell) \end{pmatrix}$$

and so, taking the determinant of both sides, we get the desired result.  $\square$

#### 4. PROOF OF THE RAMIFICATION THEOREM

We now prove our theorem.

*Proof of 1.9.* We first observe that

$$\begin{aligned} p \mid \mathrm{disc}(K) &\iff \mathrm{disc}(K) \equiv 0 \pmod{p} \\ &\iff \mathrm{disc}(\mathcal{O}_K/(p)) = \bar{0} && \text{by Lemma 3.1} \\ &\iff \prod \mathrm{disc}(\mathcal{O}_K/P_i^{e_i}) = \bar{0} && \text{by Lemma 3.2} \end{aligned}$$

Thus, if any  $e_i > 1$ , we get that  $\mathcal{O}_K/P_i^{e_i}$  has a nonzero nilpotent element by 2.1, and so  $\mathrm{disc}(\mathcal{O}_K/P_i^{e_i}) = \bar{0}$  by 2.4, thus giving  $p \mid \mathrm{disc}_{\mathbb{Z}}(\mathcal{O}_K)$  by the equivalences above.

If all  $e = 1$ , then each  $\mathcal{O}_K/P_i$  is a finite field, so  $\mathrm{disc}(\mathcal{O}_K/P_i) \neq \bar{0}$  by 2.5. Therefore, it must be that  $p \nmid \mathrm{disc}(K)$ .  $\square$

#### 5. FACTORIZATION IN QUADRATIC NUMBER FIELDS

In this section, we follow [Ash03] to determine some results about factorization of primes in quadratic number fields. First, recall the theorem

**5.1. Theorem** (Ram-Rel Identity). *Let  $A$  be an integral domain with field of fractions  $K$ ,  $L|K$  a finite separable field extension of degree  $n$ , and  $B$  the integral closure of  $A$  in  $L$ . Given a prime ideal  $P \subseteq A$ , if*

$$PB = \prod_{i=1}^g P_i^{e_i} \quad f_i = [B/P_i : A/P]$$

then

$$\sum_{i=1}^g e_i f_i = [B/PB : A/P] = n$$

Thus, for  $m \in \mathbb{Z} \setminus \{0, 1\}$ , a squarefree integer,  $\mathbb{Q}(\sqrt{m})|\mathbb{Q}$  has degree 2. Thus, for a prime  $p \in \mathbb{Z}$ , there are only three possible situations.

(a)  $g = 2, e_1 = e_2 = f_1 = f_2 = 1$ , that is,

$$(p) = P_1 P_2$$

In this situation, we say that  $p$  *splits* in  $\mathcal{O}_K$ .

(b)  $g = 1, e_1 = 1, f_1 = 2$ , that is,  $(p)$  is a prime ideal of  $\mathcal{O}_K$ . In this situations, we say that  $(p)$  is *inert*.

(c)  $g = 1, e_1 = 2, f_1 = 1$ , that is,

$$(p) = P_1^2$$

so  $p$  ramifies.

Furthermore, we will use the following result about the discriminant of  $\mathbb{Q}(\sqrt{m})$ .

**5.2. Proposition.** *The discriminant of  $\mathbb{Q}(\sqrt{m})$  is  $m$  if  $m \equiv 1 \pmod{4}$  and it is  $4m$  if  $m \equiv 2, 3 \pmod{4}$ . In particular, the discriminant is always 0 or 1 mod 4.*

*Proof.* If  $m \not\equiv 1 \pmod{4}$ ,  $\{1, \sqrt{m}\}$  is an integral basis of  $\mathbb{Q}(\sqrt{m})$ . Then,

$$\text{Tr}(a+b\sqrt{m}) = \text{tr} \begin{pmatrix} a & b \\ bm & a \end{pmatrix} = 2a \implies \text{disc}(\mathbb{Q}(\sqrt{m})) = \det \begin{pmatrix} 2 & 0 \\ 0 & 2m \end{pmatrix} = 4m$$

If  $m \equiv 1 \pmod{4}$ , then  $\{1, \frac{1+\sqrt{m}}{2}\}$  forms an integral basis and

$$\left(\frac{1+\sqrt{m}}{2}\right)^2 = \frac{m-1}{4} + \frac{1+\sqrt{m}}{2}$$

So,  $\text{Tr}(1) = 2$  and

$$\begin{aligned} \text{Tr}\left(\frac{1+\sqrt{m}}{2}\right) &= \text{tr} \begin{pmatrix} 0 & 1 \\ \frac{m-1}{4} & 1 \end{pmatrix} = 1, \\ \text{Tr}\left(\frac{m-1}{4} + \frac{1+\sqrt{m}}{2}\right) &= \text{tr} \begin{pmatrix} \frac{m-1}{4} & 1 \\ \frac{m-1}{4} & \frac{m+3}{4} \end{pmatrix} = \frac{m+1}{2} \end{aligned}$$

Thus

$$\text{disc}(\mathbb{Q}(\sqrt{m})) = \det \begin{pmatrix} 2 & 1 \\ 1 & \frac{1+m}{2} \end{pmatrix} = m$$

□

We then have the following result.

**5.3. Theorem.** *Let prime  $p \neq 2$ . Then,*

- (a)  $(p)$  ramifies as  $(p, \sqrt{m})^2$  in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \equiv 0 \pmod{p}$ .
- (b)  $(p)$  splits as  $(p) = (p, a + \sqrt{m})(p, a - \sqrt{m})$  in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \equiv a^2 \pmod{p}$  for some  $a \not\equiv 0 \pmod{p}$ .
- (c)  $(p)$  is inert in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \not\equiv a^2 \pmod{p}$  for all  $a$ .

If  $p = 2$  and  $m$  is odd, then

- (a)  $(2)$  ramifies in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \equiv 3 \pmod{4}$ .

- (b) (2) splits as  $\left(2, \frac{1+\sqrt{m}}{2}\right) \left(2, \frac{1-\sqrt{m}}{2}\right)$  in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \equiv 1 \pmod{8}$ .
- (c) (2) is inert in  $\mathbb{Q}(\sqrt{m})$  if and only if  $m \equiv 5 \pmod{8}$ .

*Proof.* We break down the various situations. Throughout, let  $D = \text{disc}(\mathbb{Q}(\sqrt{m}))$ .

- Assume  $p$  is an odd prime with  $p$  not dividing  $m$ .  $p$  does not divide the discriminant, so  $(p)$  cannot ramify.

– If  $m \equiv a^2 \pmod{p}$ ,  $a \not\equiv 0 \pmod{p}$ , then  $(p) = (p, a + \sqrt{m})(p, a - \sqrt{m})$  because

$$(p, a + \sqrt{m})(p, a - \sqrt{m}) = (p^2, pa + p\sqrt{m}, pa - p\sqrt{m}, \underbrace{a^2 - m}_{\equiv 0 \pmod{p}}) \subseteq (p)$$

and since

$$p(a + \sqrt{m} + a - \sqrt{m}) = 2ap \in (p, a + \sqrt{m})(p, a - \sqrt{m})$$

but  $a \not\equiv 0 \pmod{p}$ , so  $\gcd(2ap, p^2) = p$ , and thus  $p \in (p, a + \sqrt{m})(p, a - \sqrt{m})$ .

– If  $m \not\equiv a^2 \pmod{p}$ , then  $x^2 - m$  is irreducible  $\pmod{p}$ . Assume  $(p) = Q_1 Q_2$ . Each  $Q_i$  must have norm  $p$ , thus giving  $\mathcal{O}_K/Q_i \cong \mathbb{F}_p$ . However,  $\sqrt{m} \in \mathcal{O}_K \implies m$  has a square root in  $\mathbb{F}_p$ , a contradiction. Thus,  $(p)$  is inert.

- Let  $p$  divide  $m$ . Then,  $p$  divides the discriminant and so  $(p)$  ramifies. In fact,

$$(p, \sqrt{m})^2 = (p^2, p\sqrt{m}, m) \subseteq (p)$$

However, since  $m$  is squarefree,  $p^2 \nmid m$ , so  $\gcd(p^2, m) = p$ , so  $p \in (p, \sqrt{m})^2$ .

- Let  $p = 2$  and  $m$  be odd.
  - If  $m \equiv 3 \pmod{4} \implies D = 4m$ , then 2 divides the discriminant, so (2) ramifies. We claim  $(2) = (2, 1 + \sqrt{m})^2$ . First, we check

$$(2, 1 + \sqrt{m})^2 = (4, 2(1 + \sqrt{m}), \underbrace{1 + 2\sqrt{m} + m}_{\equiv 0 \pmod{2}}) \subseteq (2)$$

Furthermore,

$$1 + 2\sqrt{m} + m - 2(1 + \sqrt{m}) = m - 1 \equiv 2 \pmod{4}$$

so there is some  $x \in \mathbb{Z}$  such that

$$m - 1 + 4x = 2$$

thus giving us equality of ideals.

- If  $m \equiv 1 \pmod{8}$ , then  $m \equiv 1 \pmod{4}$ , so we get an integral basis  $\{1, \frac{1+\sqrt{m}}{2}\}$  and the discriminant is  $D = m$ . Therefore,  $2 \nmid D$ , so (2) does not ramify. We then compute,

$$(2, \frac{1 + \sqrt{m}}{2})(2, \frac{1 - \sqrt{m}}{2}) = (4, 1 - \sqrt{m}, 1 + \sqrt{m}, \underbrace{\frac{1 - m}{4}}_{\text{Even}}) \subseteq (2)$$



However, we also have

$$1 - \sqrt{m} + 1 + \sqrt{m} = 2 \in (2, \frac{1 + \sqrt{m}}{2})(2, \frac{1 - \sqrt{m}}{2})$$

giving us the desired ideal equality.

- If  $m \equiv 5 \pmod{8}$ , then  $m \equiv 1 \pmod{4}$ , so  $D = m$ , meaning 2 does not ramify. Consider

$$f(x) = x^2 - x + \frac{1 - m}{4} \in (\mathcal{O}_K/P)[x]$$

where  $(2) \subseteq P$  a prime ideal in  $\mathcal{O}_K$ . The roots of  $f$  are  $\frac{1 \pm \sqrt{m}}{2}$ , so  $f$  has a root in  $\mathcal{O}_K$  and hence in  $\mathcal{O}_K/P$ . However, since  $\frac{1-m}{4} \equiv 1 \pmod{2}$ ,  $f$  has no root in  $\mathbb{F}_2$ . Therefore,  $\mathcal{O}_K/P$  and  $\mathbb{F}_2$  cannot be isomorphic. If  $(2) = P_1 P_2$  in  $\mathcal{O}_K$ , then the norm of  $(2)$  is 4 and so  $P_1, P_2$  each have norm 2. Therefore,  $\mathcal{O}_K/P_i \cong \mathbb{F}_2$ , which is a contradiction. Thus,  $(2)$  must remain prime.

□

## REFERENCES

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